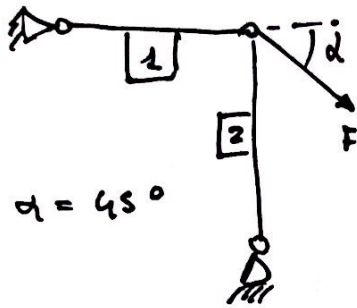


EXERCISE 2.2



Find the cross-sectional areas A_1 and A_2 such that the weight of the structure is minimized.

Constraint: yielding and buckling

The mass of the structure, m , is:

$$m = \rho L (A_1 + A_2) \sim A_1 + A_2$$

Stress inside the bars is:

$$\sigma_1 = \frac{F \cos \alpha}{A_1}$$

$$\sigma_2 = -\frac{F \sin \alpha}{A_2}$$

$$= \frac{\sqrt{2}}{2} \frac{F}{A_1}$$

$$= -\frac{\sqrt{2}}{2} \frac{F}{A_2}$$

To avoid yielding, one should ones that

$$|\sigma| \leq \sigma_0 \rightarrow \begin{aligned} \sigma_1 &\leq \sigma_0 \\ \sigma_2 &\geq -\sigma_0 \end{aligned}$$

$$\frac{\sqrt{2}}{2} \frac{F}{A_1} \leq \sigma_0$$

$$-\frac{\sqrt{2}}{2} \frac{F}{A_2} \geq -\sigma_0$$

$$\boxed{\begin{aligned} A_1 &\geq \frac{\sqrt{2} F}{2 \sigma_0} \\ A_2 &\geq \frac{\sqrt{2} F}{2 \sigma_0} \end{aligned}}$$

Now, let us consider buckling:

From theory it is known that the Eulerian critical load for a bar subjected to compressive stress is

PIN BAR $\rightarrow P_c = \pi^2 \frac{EI}{L^2}$



Therefore, the constraint becomes, considering a safety factor against buckling equal to 4:

$$\frac{P_c}{4} \geq \sigma_2 A_2 = \frac{F}{\sqrt{2}}$$

$$A_2^2 \geq \frac{16FL^2}{\sqrt{2}\pi E}$$

The problem is then:

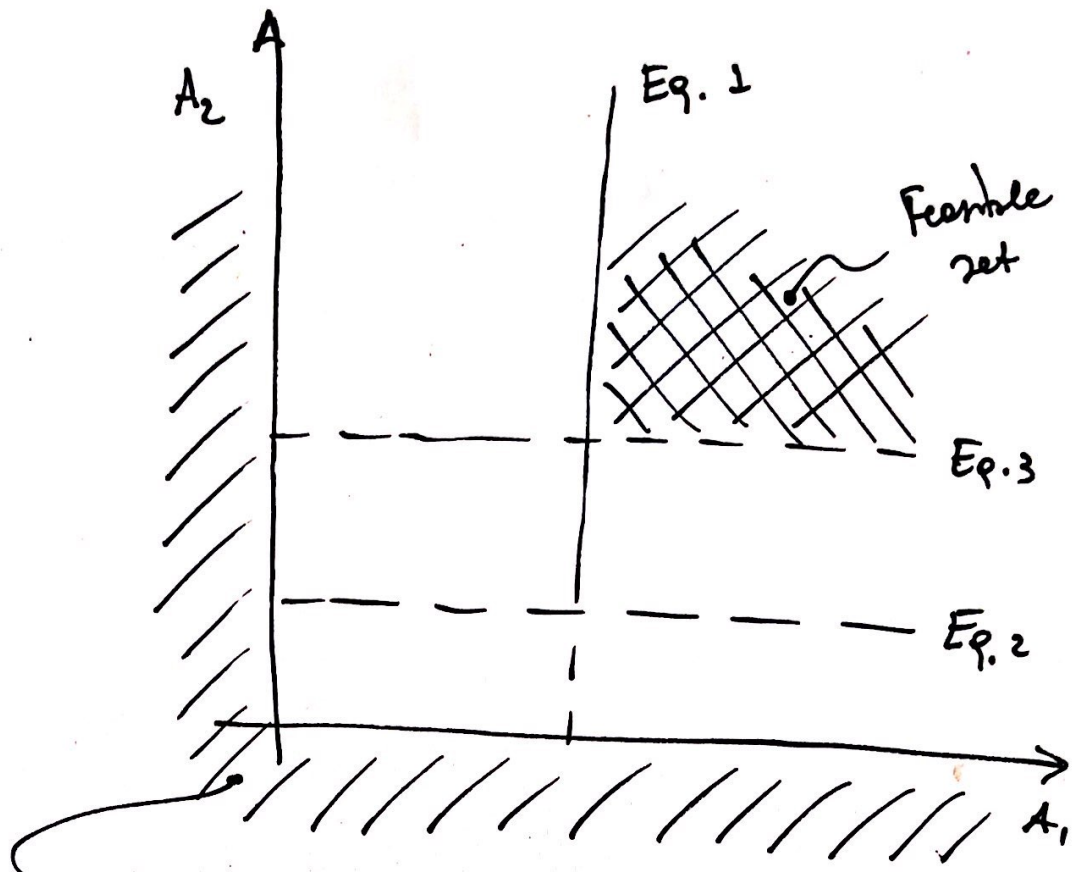
$$\begin{cases} \text{min } A_1 + A_2 \\ \text{s.t. } \begin{cases} A_1 \geq F/\sqrt{2}\sigma_0 & \leftarrow \text{YIELDING} \\ A_1 \geq F/\sqrt{2}\sigma_0 & \leftarrow \text{YIELDING} \\ A_2^2 \geq 16FL^2/\sqrt{2}\pi E & \leftarrow \text{BUCKLING} \end{cases} \end{cases}$$

NOTE: Note that elastic buckling can happen only to bars subjected to compressive axial loads.

Constraint against buckling for bar 1 has not any sense.

(2)

Activity of constraints:



Areas are always greater than zero.

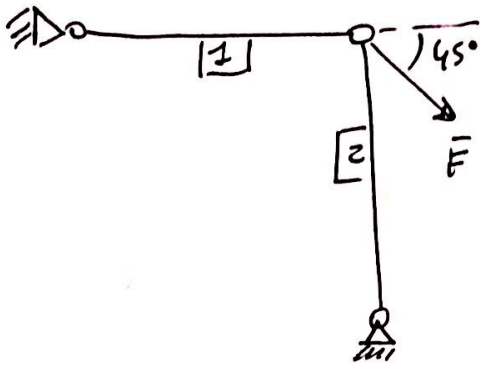
$$\begin{cases} A_1 \geq F/\sqrt{2}\sigma_0 & \text{Eq. 1} \\ A_2 \geq F/\sqrt{2}\sigma_0 & \text{Eq. 2} \\ A_2^2 \geq 16FL^2/\sqrt{2}\pi E & \text{Eq. 3} \end{cases}$$

In this representation, Eq. 2 is not an active constraint, Eq. 3 is an active constraint.

$$\text{Eq. 2} = \text{Eq. 3} \quad \text{iff} \quad \sigma_0^2 = \frac{\sqrt{2}}{32} \frac{F\pi E}{L^2}$$

(3)

Exercise 2.2 bis



Find the cross sectional areas A_1 and A_2 to maximize the stiffness of the structure.

The volume of the structure cannot exceed a value V_0 .

Moreover, consider that bars must not yield.

The stiffness is the reciprocal of the compliance: $K = \frac{1}{G}$

(This is true for linear elasticity. For visco-elasticity, i.e. polymers, relationships are way more complex).

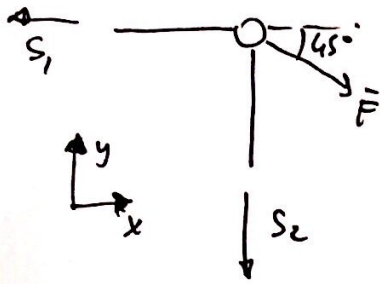
In order to maximize the stiffness we should minimize the compliance

max STIFFNESS $\longleftrightarrow$ min COMPLIANCE

Let define the compliance as the displacement of the free node:

$$G \sim |\underline{\mu}| = \sqrt{\mu_x^2 + \mu_y^2}$$

In order to find the expansion of the displacements one should calculate forces inside elements, then stresses, then strains and eventually the elongations.



$$F_x - S_1 = 0$$

$$F_x = F \frac{\sqrt{2}}{2}$$

$$F_y - S_2 = 0$$

$$F_y = -F \frac{\sqrt{2}}{2}$$

$$\begin{bmatrix} F_x \\ F_y \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} S_1 \\ S_2 \end{bmatrix}$$

$$\underline{F} = \underline{B}^T \underline{S}$$

$$\underline{B}^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} S_1 \\ S_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} F_x \\ F_y \end{bmatrix}$$

$$S_1 = F \sqrt{2}/2$$

$$S_2 = -F \sqrt{2}/2$$

The matrix $\underline{B}^T$ is important because it is allow to compute displacements too!

Principle of virtual works: work performed by external forces must be equal to the work done by internal forces.

$$\underline{F}^T \cdot \underline{\mu} = \underline{S}^T \cdot \underline{\delta}$$

$$\text{but: } \underline{F} = \underline{B}^T \underline{S}$$

$$\text{so: } \underline{F}^T = \underline{S}^T \underline{B}$$

$$(\underline{S}^T \underline{B}) \cdot \underline{\mu} = \underline{S}^T \cdot \underline{\delta}$$



$$\underline{B} \cdot \underline{\mu} = \underline{\delta}$$

$\underline{\delta}$ vector of elongations,

$\underline{\mu}$ vector of displacements

So, the displacements are:

$$\underline{\mu} = \underline{B}^{-1} \underline{\delta}$$

$$\underline{B}^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\delta_1 = \frac{\sigma_1}{E} L_1 = \frac{F\sqrt{2}}{2A_1} \frac{L_1}{E}$$

$$\delta_2 = -\frac{F\sqrt{2}}{2A_2} \frac{L_2}{E}$$

$$\underline{\mu} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{F\sqrt{2}}{2A_1} L_1/E \\ -\frac{F\sqrt{2}}{2A_2} L_2/E \end{bmatrix} \quad \mu_x = \frac{\sqrt{2}}{2} \frac{FL_1}{A_1 E}$$

$$\mu_y = -\frac{\sqrt{2}}{2} \frac{FL_2}{A_2 E}$$

The compliance is then:

$$C = \sqrt{\frac{1}{2} \frac{F^2 L_1^2}{E^2 A_1^2} + \frac{1}{2} \frac{F^2 L_2^2}{E^2 A_2^2}} = \frac{1}{\sqrt{2}} \frac{F}{E} \cdot \sqrt{\frac{L_1^2}{A_1^2} + \frac{L_2^2}{A_2^2}}$$

The volume is:

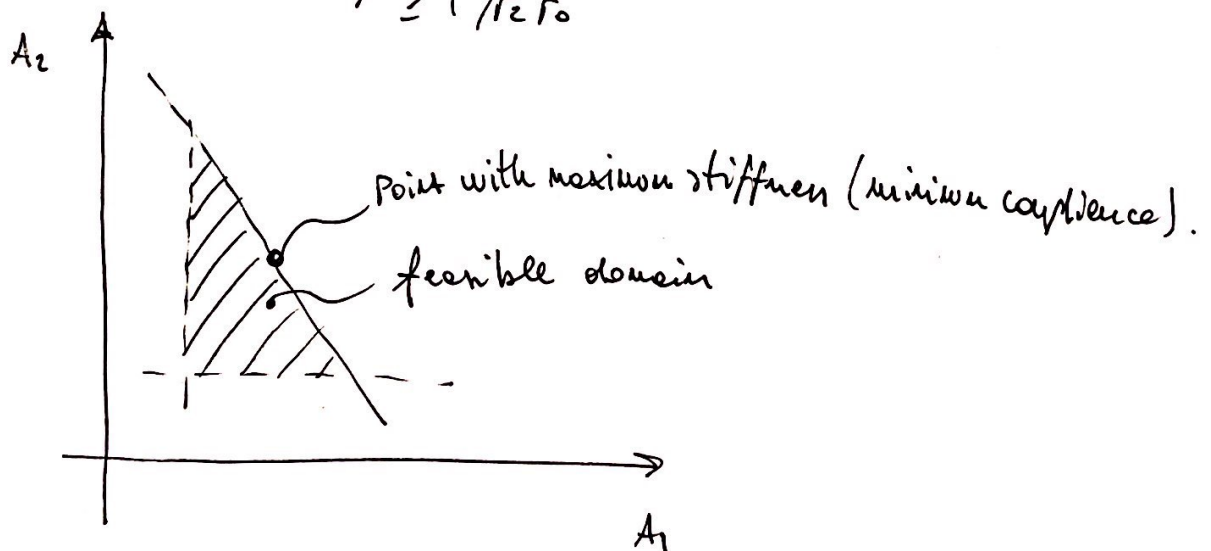
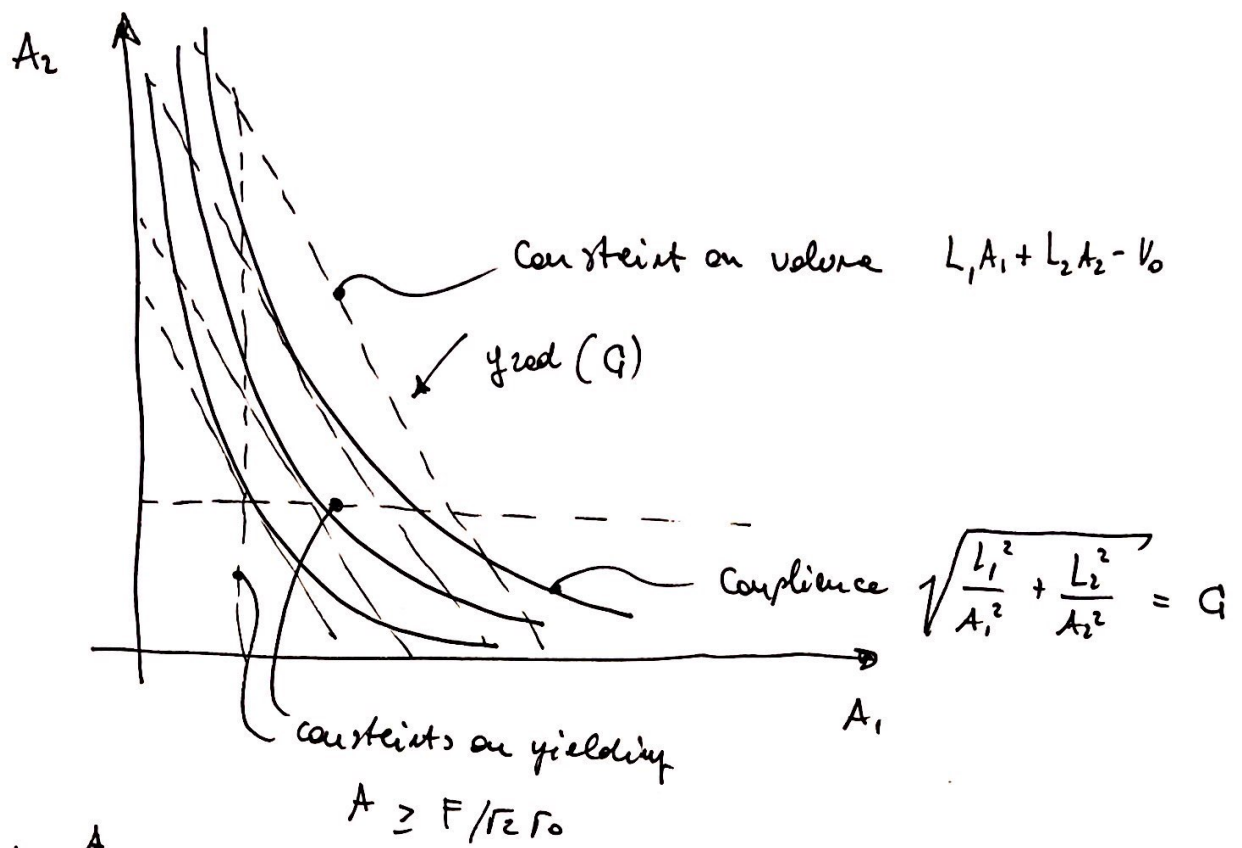
$$V = L_1 A_1 + L_2 A_2 \leq V_0$$

$$L_1 A_1 + L_2 A_2 - V_0 \leq 0$$

Therefore the optimization problem is

$$\left\{ \begin{array}{l} \min \sqrt{\frac{L_1^2}{A_1^2} + \frac{L_2^2}{A_2^2}} \\ \text{s.t.} \left\{ \begin{array}{l} L_1 A_1 + L_2 A_2 - V_0 \leq 0 \\ A_1 \geq F/\sqrt{2} \sigma_0 \\ A_2 \geq F/\sqrt{2} \sigma_0 \end{array} \right. \end{array} \right.$$

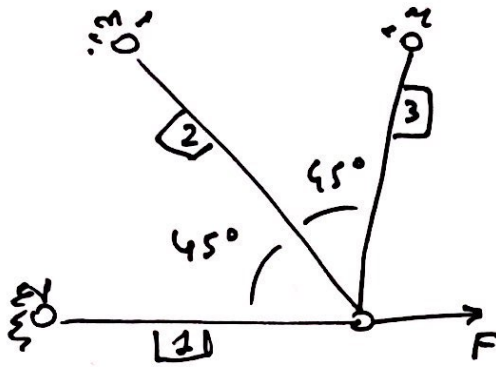
(3)



For questions, contact :

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EXERCISE 2.5



Minimize mass under stress constraints.

$$l_1 = l_2 = l_3 = L$$

The mass of the structure, that is the objective function, is:

$$m = L (\rho_1 A_1 + \rho_2 A_2 + \rho_3 A_3)$$

The stress constraints are:

$$|\sigma_1| \leq \sigma_0 \quad |\sigma_2| \leq \sigma_0 \quad |\sigma_3| \leq \sigma_0$$

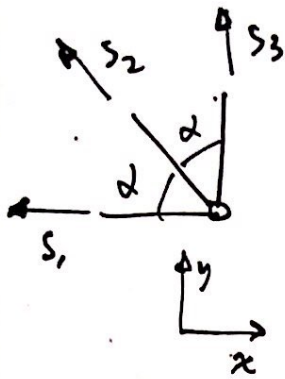
In order to solve the problem one should express the constraints as an explicit function of the design variables

$$\underline{DV} = \begin{Bmatrix} A_1 \\ A_2 \\ A_3 \end{Bmatrix} \quad \underline{\text{aim:}} \quad \text{to find } \sigma_i(A_1, A_2, A_3) \leq \sigma_0 \quad i = 1, 2, 3$$

Note that the structure is hyperstatic, therefore the distribution of forces inside its elements follows their stiffness and ensures compatibility of displacements.

(1)

As usual, write the equilibrium of forces around the free node:



$$F_x - S_1 - S_2 \frac{\sqrt{2}}{2} = 0$$

$$F_y + S_2 \frac{\sqrt{2}}{2} + S_3 = 0$$

$$\underline{F} = \underline{B}^T \underline{S}$$

$$\underline{F} = \underbrace{\begin{bmatrix} 1 & +\sqrt{2}/2 & 0 \\ 0 & -\sqrt{2}/2 & -1 \end{bmatrix}}_{\underline{B}^T} \begin{bmatrix} S_1 \\ S_2 \\ S_3 \end{bmatrix}$$

This system cannot be solved.

The compatibility condition, must be kept into consideration. If s is the force inside the bar, then:

$$S = \sigma \cdot A = E \cdot \varepsilon \cdot A = \frac{E}{L} \delta A$$

$$\boxed{K = \frac{EA}{L}}$$

This is the stiffness of a bar

Therefore:

$$\underline{\varepsilon} = \underbrace{\begin{bmatrix} \frac{EA_1}{L} & 0 & 0 \\ 0 & \frac{EA_2}{L} & 0 \\ 0 & 0 & \frac{EA_3}{L} \end{bmatrix}}_{\underline{D}} \underbrace{\begin{bmatrix} \delta_1 \\ \delta_2 \\ \delta_3 \end{bmatrix}}_{\underline{\delta}}$$

$$\underline{\varepsilon} = \underline{D} \underline{\delta}$$

②

We now have two relationships:

$$\underline{\underline{F}} = \underline{\underline{B}}^T \underline{\underline{s}} \quad \text{and} \quad \underline{\underline{s}} = \underline{\underline{D}} \underline{\underline{\epsilon}}$$

From the principle of virtual work, we have:

$$\underline{\underline{\delta}} = \underline{\underline{B}} \underline{\underline{\mu}} \quad \left(\text{It comes from } \underline{\underline{F}}^T \underline{\underline{\mu}} = \underline{\underline{s}}^T \underline{\underline{\epsilon}} \right)$$

Therefore:

$$\underline{\underline{s}} = \underline{\underline{D}} \underline{\underline{\delta}} = \underline{\underline{D}} \underline{\underline{B}} \underline{\underline{\mu}}$$

$$\underline{\underline{F}} = \underline{\underline{B}}^T \underline{\underline{s}} = \underline{\underline{B}}^T \underline{\underline{D}} \underline{\underline{B}} \underline{\underline{\mu}}$$

$$\boxed{\underline{\underline{K}} = \underline{\underline{B}}^T \underline{\underline{D}} \underline{\underline{B}}}$$

STIFFNESS MATRIX OF
THE STRUCTURE

By performing some calculations:

$$\underline{\underline{K}} = \begin{bmatrix} A_1 + \frac{A_2}{2} & -\frac{A_2}{2} \\ -\frac{A_2}{2} & A_1 + \frac{A_2}{2} \end{bmatrix} \cdot \frac{E}{L}$$

Solving $\underline{\underline{F}} = \underline{\underline{K}} \underline{\underline{\mu}}$ for $\underline{\underline{\mu}} = \begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix}$ leads:

$$\mu_x = \frac{FL}{EA_1} \left(\frac{2A_1 + A_2}{2A_2 + 2A_1} \right)$$

$$\mu_y = \frac{FL}{EA_1} \left(\frac{A_2}{2A_1 + 2A_2} \right)$$

③

In order to calculate stress inside bars, a matrix $\underline{A}$ is defined :

$$\underline{A} = \begin{bmatrix} \frac{1}{A_1} & 0 & 0 \\ 0 & \frac{1}{A_2} & 0 \\ 0 & 0 & \frac{1}{A_3} \end{bmatrix}$$

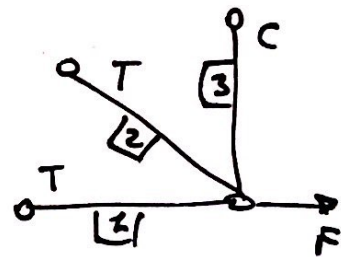
Thvs : $\underline{\sigma} = \underline{A} \cdot \underline{\epsilon} = \underline{A} \underline{D} \underline{B} \underline{\mu}$.

After some calculations :

$$\sigma_1 = \frac{F}{2A_1 + 2A_2} \left(2 + \frac{A_2}{A_1} \right) \quad \text{Tension}$$

$$\sigma_2 = \frac{\sqrt{2} F}{2A_1 + 2A_2} \quad \text{Tension}$$

$$\sigma_3 = - \frac{F A_2 / A_1}{2A_1 + 2A_2} \quad \text{Compression}$$



So constraints become :

$$\sigma_1 \leq \sigma_0 \quad \frac{F(2A_1 + A_2)}{2A_1(A_1 + A_2)} \leq \sigma_0$$

$$\sigma_2 \leq \sigma_0 \quad \frac{F}{\sqrt{2}(A_1 + A_2)} \leq \sigma_0$$

$$\sigma_3 \geq -\sigma_0 \quad \frac{F A_2}{2A_1(A_1 + A_2)} \leq \sigma_0$$

(4)

The optimization problem can be formulated:

$$\begin{cases} \min & L(p_1 A_1 + p_2 A_2 + p_3 A_3) \\ \text{s.t.} & \begin{cases} \frac{F(2A_1 + A_2)}{2A_1(A_1 + A_2)} - \sigma_0 \leq 0 \\ \frac{F}{\sqrt{2}(A_1 + A_2)} - \sigma_0 \leq 0 \\ \frac{FA_2}{2A_1(A_1 + A_2)} - \sigma_0 \leq 0 \end{cases} \end{cases}$$

This approach can be used for simple problems where everything can be expressed as function of the design variables.